

Select AI Proxy Integration

Oracle Autonomous AI Database with OCI PostgreSQL

What is Select AI?

Use natural language to interact with your database and LLMs through SQL to enhance user productivity and develop AI-based applications. Select AI simplifies and automates using generative AI, whether generating, running, and explaining SQL from a natural language prompt, using retrieval augmented generation with vector stores, generating synthetic data, or chatting with the LLM.

When you use Select AI, Autonomous AI Database manages the process of converting natural language into SQL. This means you can provide a natural language prompt instead of SQL code to interact with your data. Select AI serves as a productivity tool for SQL users and developers and enables non-expert SQL users to derive useful insights from their data, without having to understand data structures or technical languages.

[\[1\]](#)

What is Select AI Proxy Integration?

Select AI runs natively inside Oracle Autonomous AI Database and Oracle AI Database, both of which can operate as an AI proxy database, also referred to as "sidecar". An AI proxy database can support both local and external data sources (on-premises, cloud, or third-party). Using standard Oracle federation mechanisms such as Database Links, Cloud Links, Table Hyperlinks, and Federated Tables, Select AI generates federated SQL from natural language prompts using metadata across Oracle and non-Oracle systems.

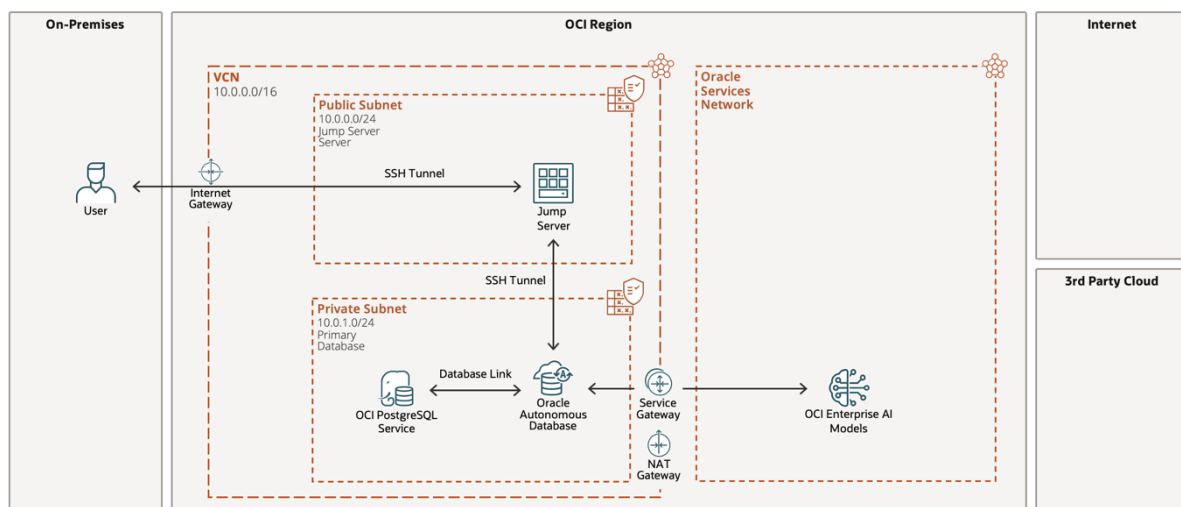
Autonomous AI Database hosts act as a central metadata and processing layer for both local and external data sources. The AI proxy database controls distributed query processing while external systems remain authoritative for their data.

An AI proxy database is an Autonomous AI Database instance that runs Select AI on behalf of local or external data sources. It does not contain the external data. Instead, it uses metadata exposed through local database objects (tables and views) that reference remote data sources such as views defined on Database Links or Cloud Links, External Tables over Table Hyperlinks, and Federated Tables to interpret natural language requests and generate SQL that runs across distributed systems. The AI proxy database can also contain local data in its own schema. [\[2\]](#)

Select AI Proxy Integration Scenario

In this technical guide we will be taking a look at how to provision the OCI Services and configure the Select AI Proxy Integration between an Oracle Autonomous AI Database and an OCI PostgreSQL Database. Each database will host a dataset where we will execute a single Natural Language query from the Autonomous AI Database which will query across both data sources while leaving data in place through the Select AI Proxy Integration.

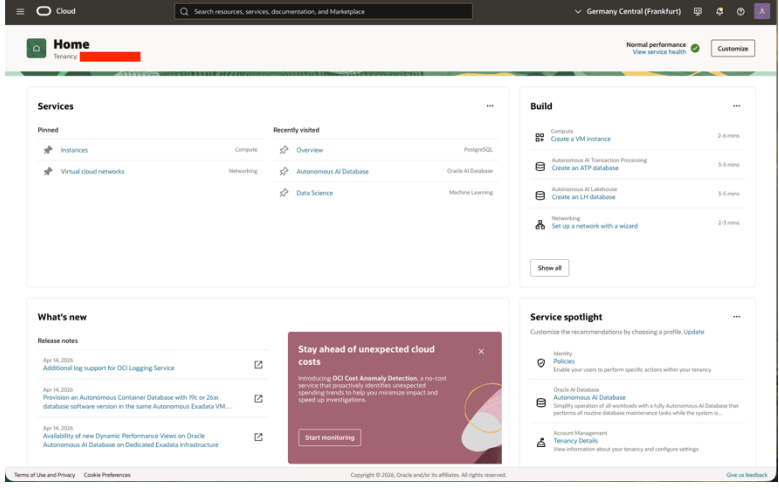
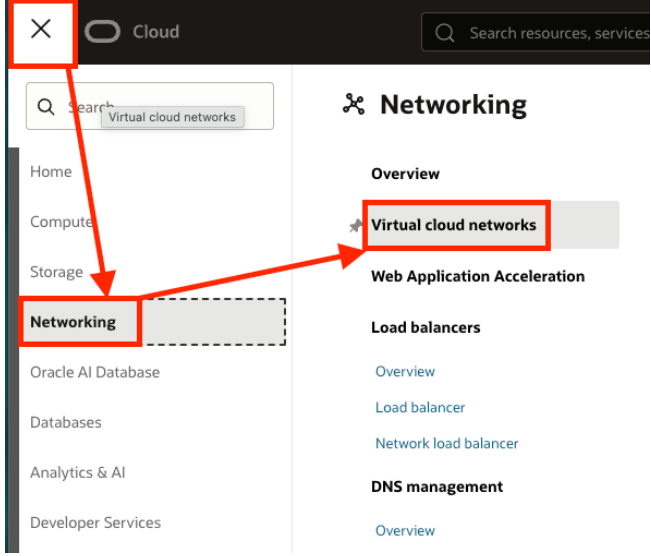
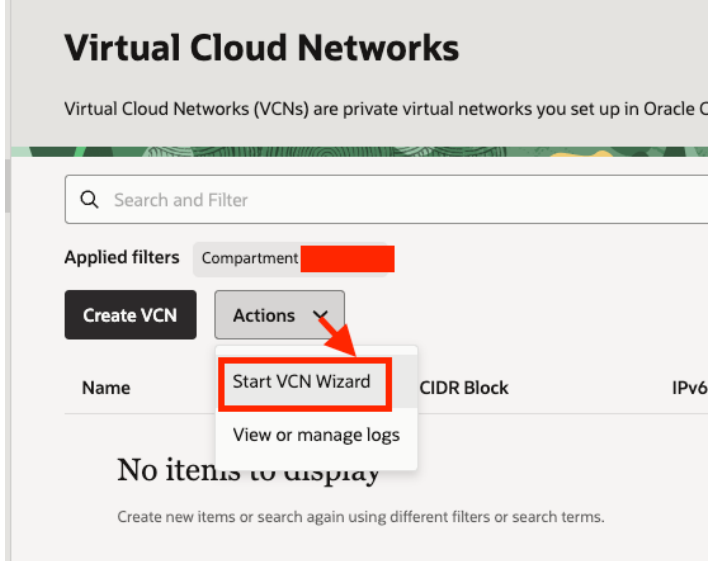
Select AI Proxy Integration Scenario Architecture



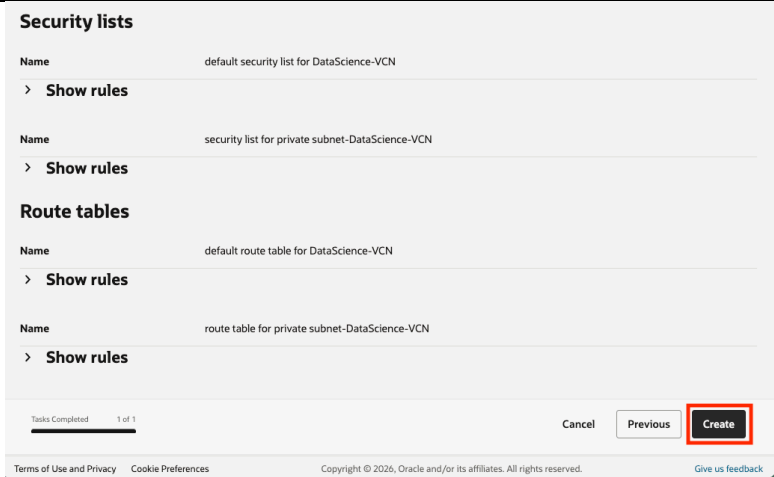
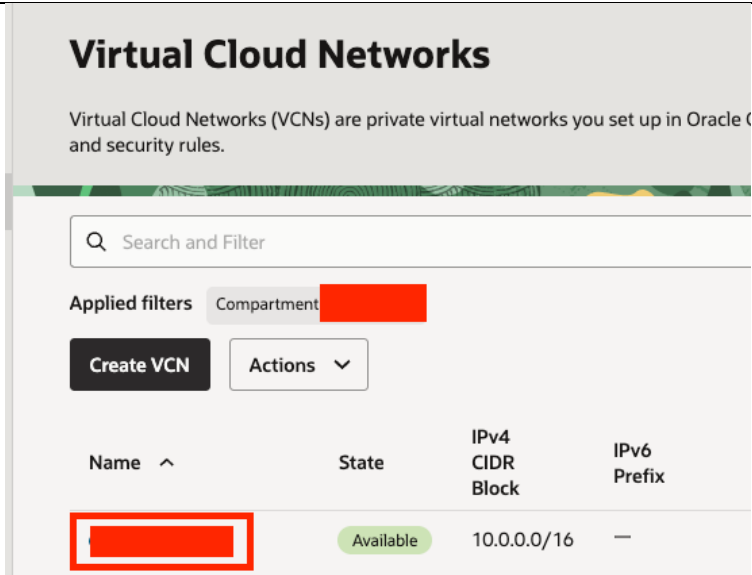
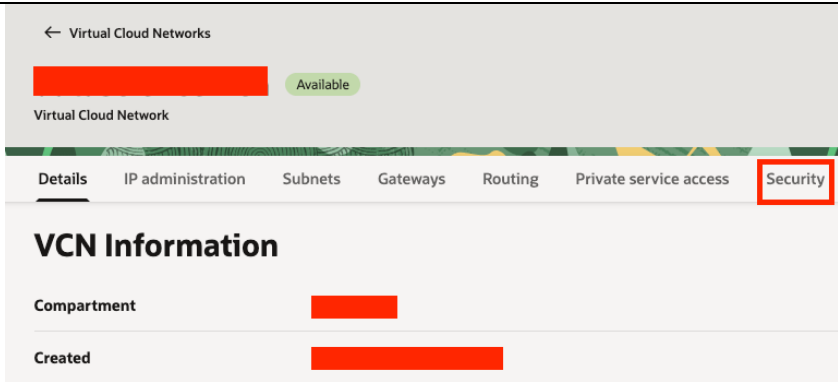
Select AI Proxy Integration Pre-Requisites

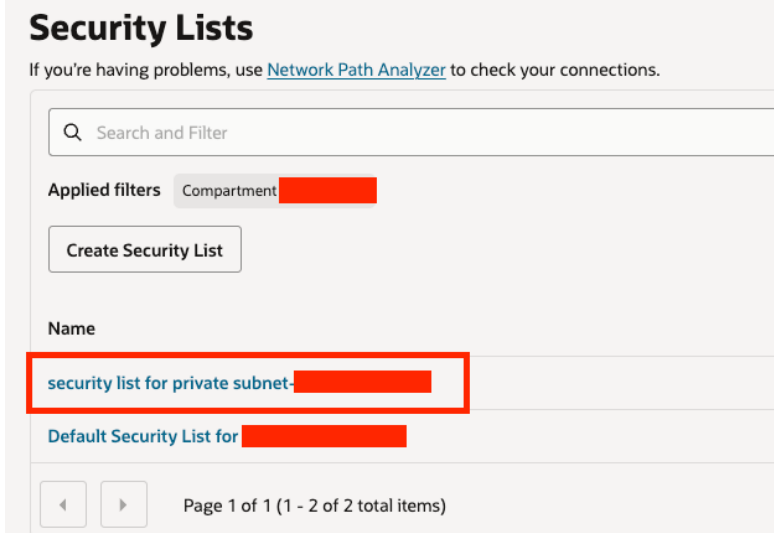
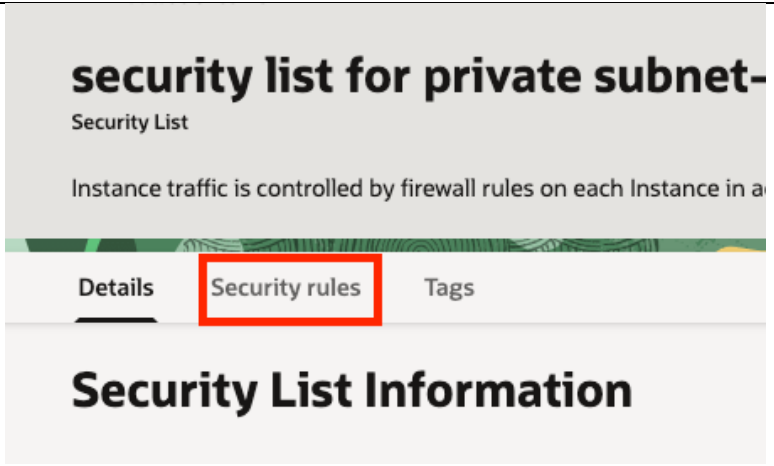
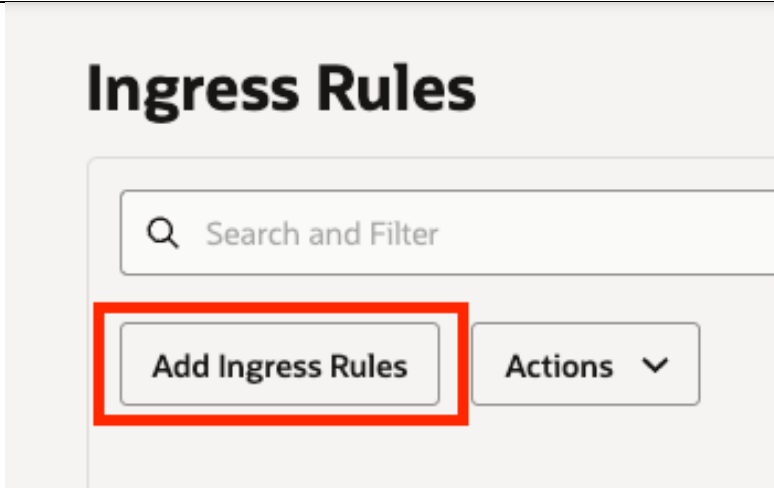
- OCI Cloud account.
- Permissions to create and manage all resources within a compartment.
- SQL Developer Desktop or SQL Developer VS Code Extension Installed.
- Following Resources:
 - GitHub Directory – **07-getting-started-proxy-integration/**
 - 00 – Connect PostgreSQL – Anon.ipynb
 - 07 – Demo Select AI Proxy Integration – Anon.sql
 - SALES.csv
 - GitHub Directory – **00-getting-started-setup-guide/**
 - 00 – Setup Select AI – Anon.sql

Select AI Proxy Integration Technical Guide

Step	Screenshot
Login to the OCI Console. https://www.cloud.oracle.com	
First, we will create a Virtual Cloud Network to host our Databases. OCI Menu > Networking > Virtual Cloud Network.	
From Actions menu select Start VCN Wizard.	

Select Create VCN with Internet Connectivity. Select Start VCN Wizard.	Start VCN Wizard To make it easier to set up a virtual cloud network (VCN) and connect to it, the Console has the following wizards that walk you through network setup. Connection Type Create VCN with Internet Connectivity Creates a VCN with a public subnet that can be reached from the internet. Also creates a private subnet that can connect to the internet through a NAT gateway, and also privately connect to the Oracle Services Network. Includes: VCN, public subnet, private subnet, internet gateway (IG), NAT gateway (NAT), service gateway (SG). Add Internet Connectivity and Site-to-Site VPN to a VCN Adds a Site-to-Site VPN between your on-premises network and a VCN you select. If the VCN has a public subnet an Internet Gateway is also added. Includes: VCN, public subnet, private subnet, dynamic routing gateway (DRG), virtual customer-premises equipment (CPE), Site-to-Site VPN, internet gateway (IG). Cancel Start VCN Wizard
Enter your VCN Name. Select your Compartment. Configure a VCN CIDR Block. I left as the default: 10.0.0.0/16 Please consult your Networking team for best practises.	Basic information VCN name Compartment Configure VCN VCN IPv4 CIDR block 10.0.0.0/16
Configure you Public and Private Subnet CIDR Block. I left as the defaults. Public: 10.0.0.0/24 Private: 10.0.1.0/24 Please consult your Networking team for best practises.	Configure public subnet IP address type IPv4 CIDR block IPv4 CIDR block 10.0.0.0/24 Maximum item count reached + Another IP address type Configure private subnet IP address type IPv4 CIDR block IPv4 CIDR block 10.0.1.0/24 Maximum item count reached + Another IP address type

Click Next to be taken to the Review Page. Click Create. Your VCN will be created.	
We will now add rules to our Security Lists to allow network traffic to flow between subnets as our databases will be placed in the Private Subnet. Select your VCN.	
Select the Security Tab.	

Select the Security List for our Private Subnet.	
Select Security Rules.	
Select Add Ingress Rules.	

Create two Ingress Rules.

Rule 1

Source Type: CIDR

Source CIDR:

10.0.0.0/16

IP Protocol: TCP

Destination Port: 1522

Rule 2

Source Type: CIDR

Source CIDR:

10.0.0.0/16

IP Protocol: TCP

Destination Port: 5432

Click **Add Ingress Rules**.

Please consult your Networking team for best practises.

Ingress Rule 1

Allows TCP traffic for ports: all

Stateless

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. Learn about [stateful](#) and [stateless](#) rules.

Source Type: CIDR

Source CIDR: 10.0.0.0/16

IP Protocol: TCP

Source Port Range:

Destination Port Range: 1522

Description: Autonomous

Ingress Rule 2

Allows TCP traffic for ports: all

Stateless

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. Learn about [stateful](#) and [stateless](#) rules.

Source Type: CIDR

Source CIDR: 10.0.0.0/16

IP Protocol: TCP

Source Port Range:

Destination Port Range: 5432

Description: PostgreSQL

Cancel Add Ingress Rules

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We will then create a Network Security Group to control the traffic coming in/out of our Autonomous AI Database.

Scroll down and Click **Create Network Security Group**.

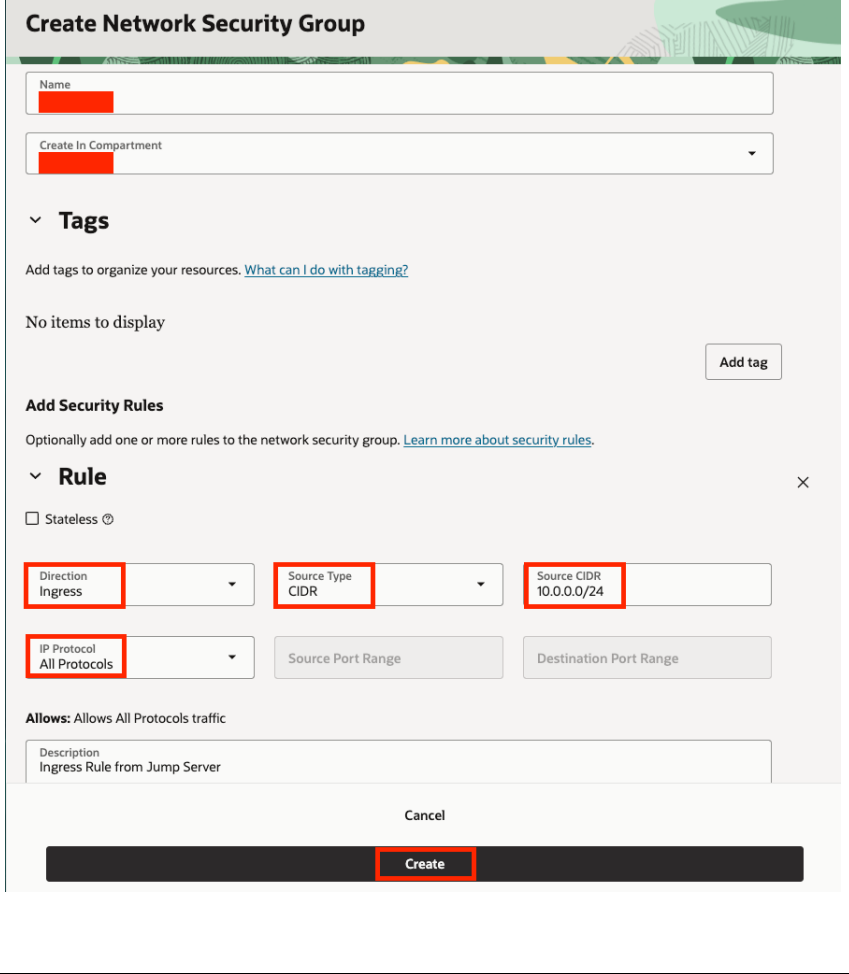
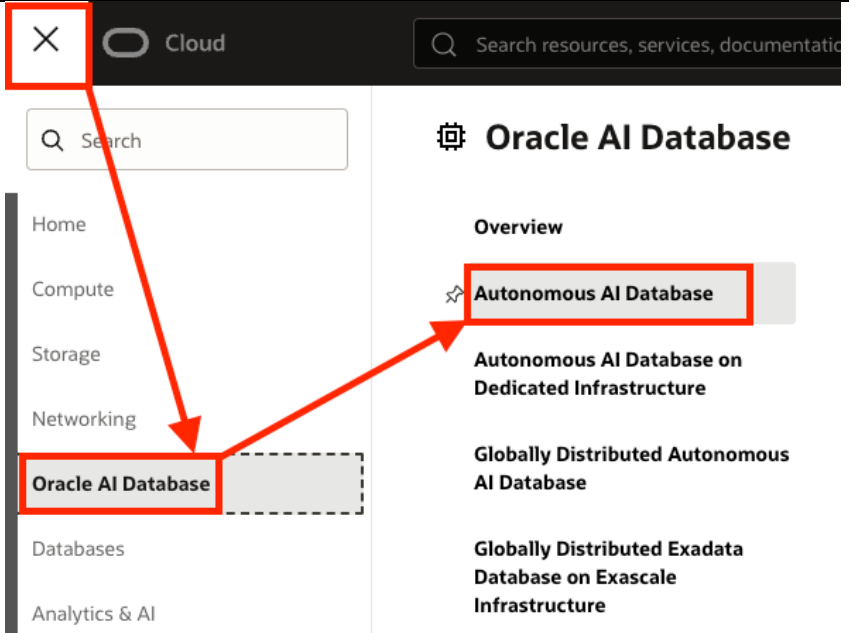
Network Security Groups

A network security group consists of a set of VNICs and a set of security rules that define the security posture. For comparison, a VCN security list consists of a set of security [groups](#).

Search and Filter

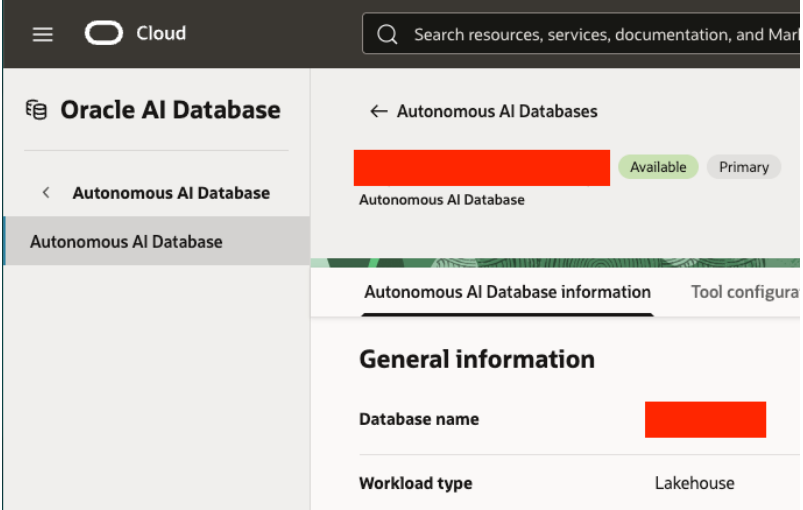
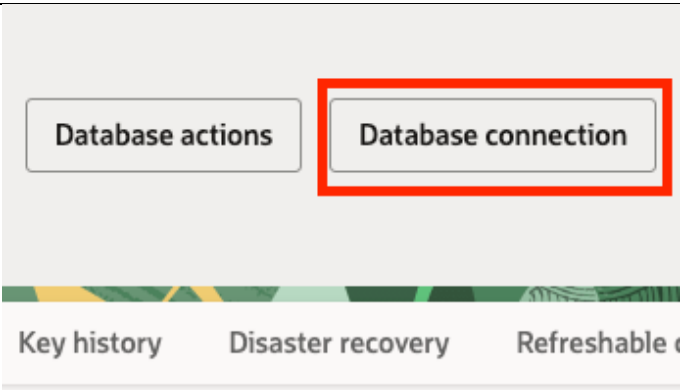
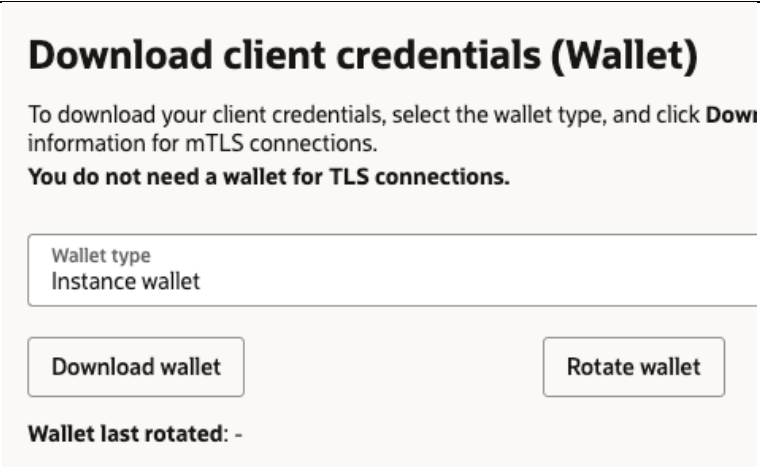
Applied filters: Compartment

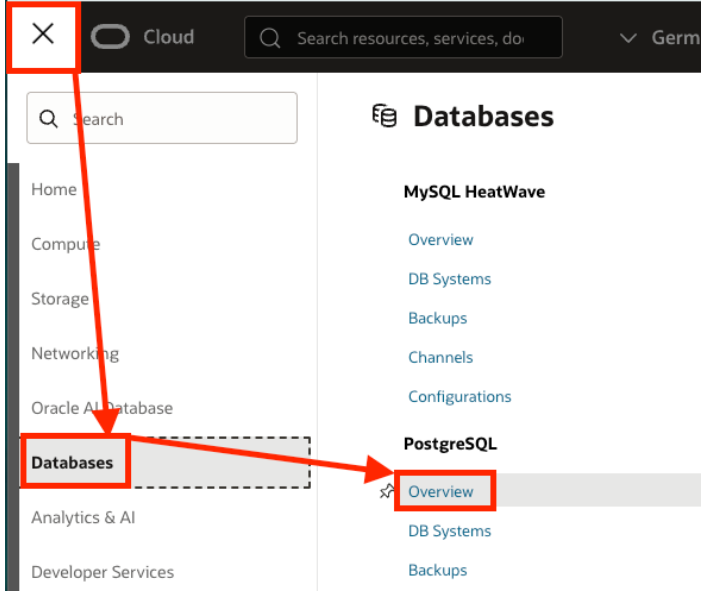
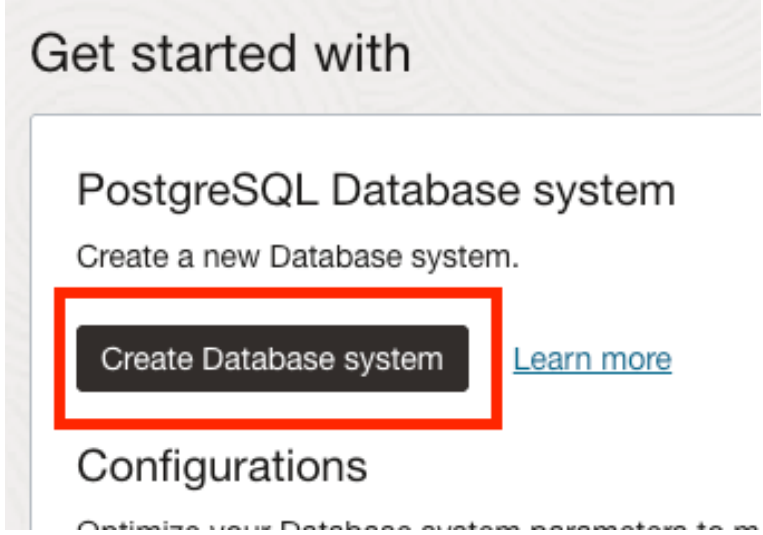
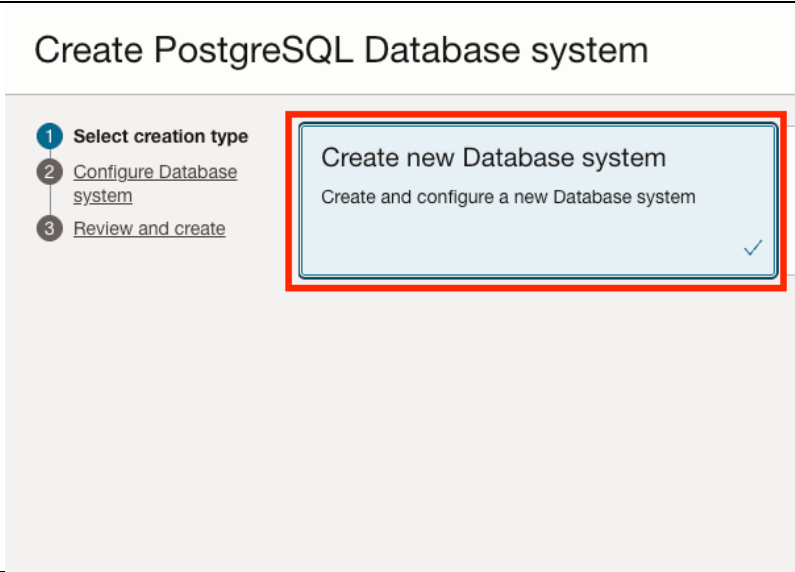
Create Network Security Group

Give it a Name. Select your Compartment. Add an Ingress Rule: Source Type: CIDR Source CIDR: 10.0.0.0/24 IP Protocol: All Destination Port: 5432 This will allow all traffic from our public subnet into our Autonomous AI Database Click Create. Please consult your Networking team for best practises.	
We will now provision our Autonomous AI Database. OCI Menu > Oracle AI Database > Autonomous AI Database.	

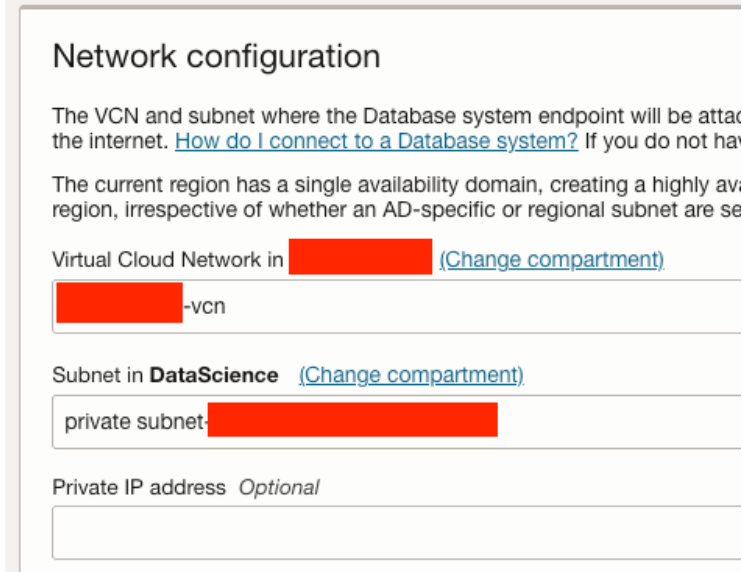
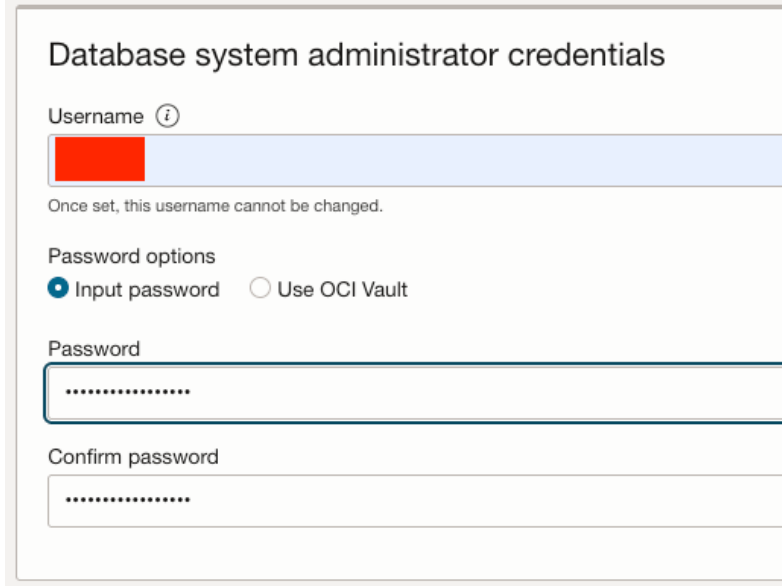
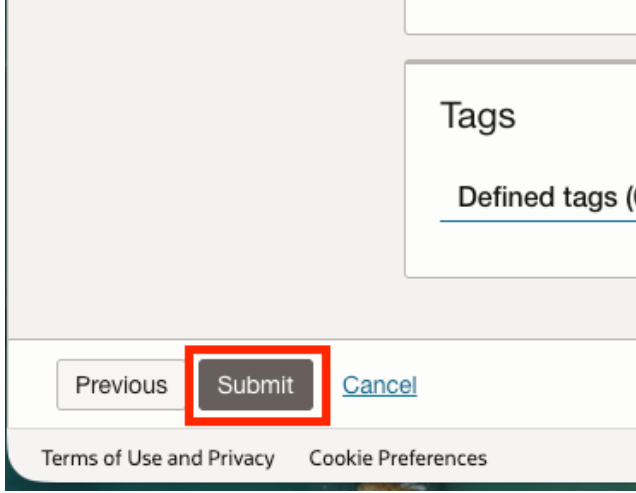
Click Create Autonomous AI Database.	Autonomous AI Databases Autonomous AI Database delivers fast performance and requires n more Search and Filter Applied filters Compartment Create Autonomous AI Database
Enter Display Name. Enter Database Name. Select your Compartment. Select Workload Type as Lakehouse.	Display name Database name Compartment Workload type Lakehouse Built for analytics and AI. Fast insights from a single lakehouse for all your data. Transaction Processing Built for transactional workloads. High concurrency for short-running queries and transactions.
Select Database Version as 26ai. Select ECPU Count as 2. Keep Auto Scaling turned on. Select Storage as 1 TB (default).	An Autonomous AI Database for Developers is a low-cost, fixed-shape instance intended for production, you may upgrade your developer database to a full paid database that can be s Choose database version 26ai ECPU count 2 Compute auto scaling ☒ Allows system to expand up to three times the specified ECPU count as demand increases. Storage (TB) 1

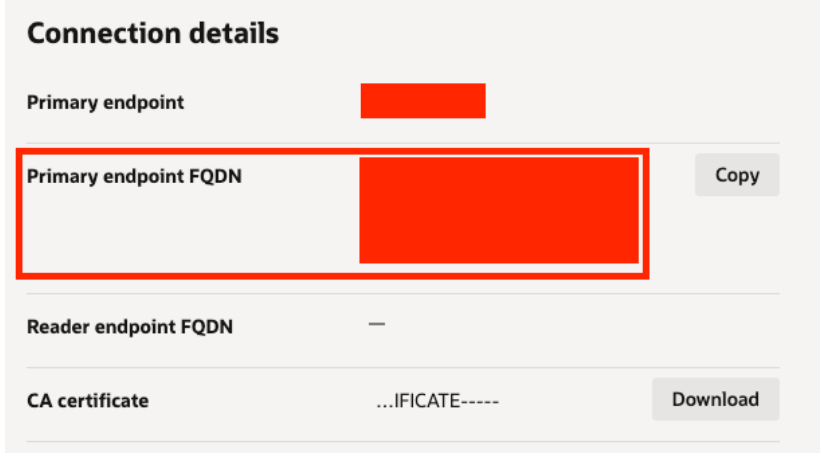
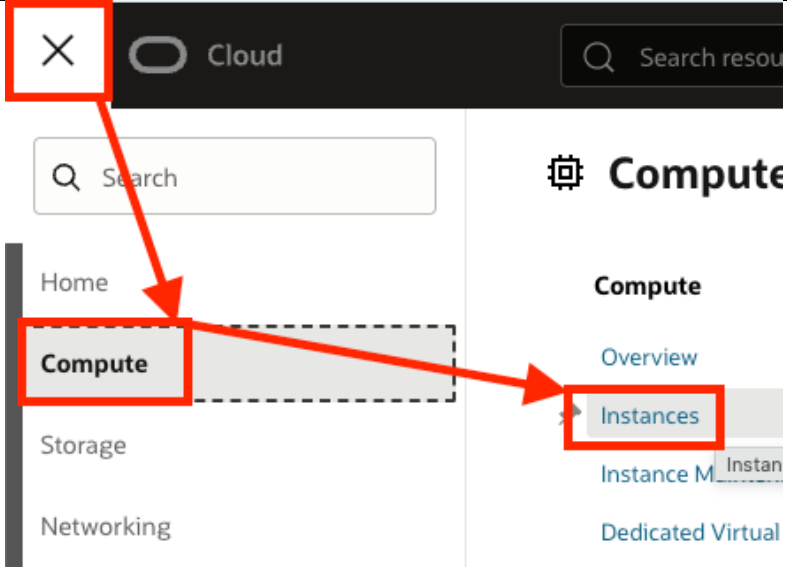
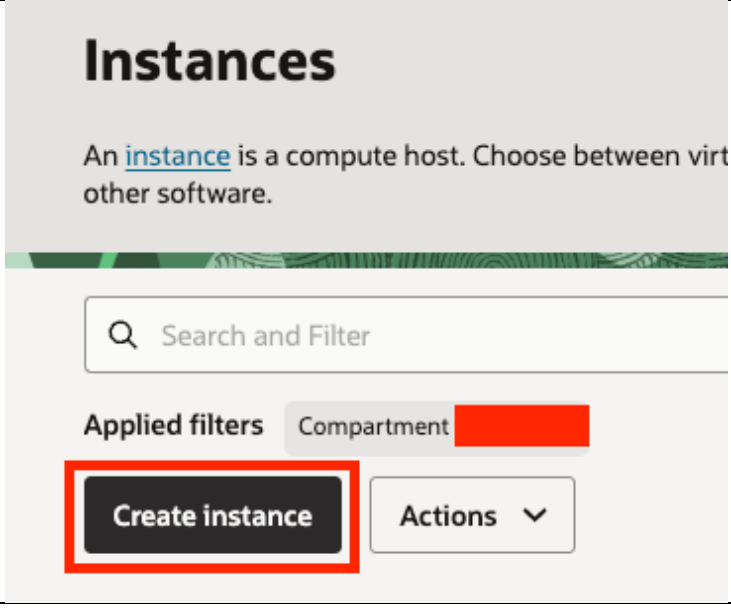
Enter and Confirm ADMIN schema password.	Administrator credentials creation Set the password for your Autonomous AI Database ADMIN user here. Username ADMIN Password Confirm password
Select Access Type as Private Endpoint Access Only. Select the VCN we created earlier. Select the Private Subnet we created earlier.	Network access Access type Secure access from everywhere Allow users with database credentials to access the database from the internet. Secure access from allowed IPs and VCNs only Restrict access to specified IP addresses and VCNs. Private endpoint access only Restrict access to a private endpoint within an OCI VCN. Compartment [redacted] Virtual cloud network [redacted] Compartment [redacted] Subnet [redacted]
Expand the Advanced Options section. Add the Network Security Group that we created earlier. Click Create.	Advanced options Private endpoint access Private IP address Host name prefix Network security groups (NSGs) An NSG has a set of security rules that control allowed types of inbound and outbound traffic. Learn more Compartment [redacted] Network security groups [redacted] × + Another network security group Allow public access ☐

Your Autonomous AI Database will take a couple minutes to provision and showcase the status as Available. Make note of the Private IP Address of the Database.	 The screenshot shows the Oracle Cloud console interface. At the top, there's a search bar and a navigation menu. The main content area displays 'Oracle AI Database' with a sub-section for 'Autonomous AI Database'. A red box highlights the 'Available' status. Below this, there's a 'General information' section showing the 'Database name' and 'Workload type' (Lakehouse).
We can now download our Autonomous AI Database Wallet File. Click Database Connection.	 The screenshot shows the 'Database actions' section of the Oracle Cloud console. A red box highlights the 'Database connection' button. Below this, there are links for 'Key history', 'Disaster recovery', and 'Refreshable'.
Click Download Wallet. Enter a Password. Click Download.	 The screenshot shows the 'Download client credentials (Wallet)' page. It includes instructions on how to download the wallet, a 'Wallet type' dropdown set to 'Instance wallet', and buttons for 'Download wallet' and 'Rotate wallet'. A 'Wallet last rotated' field is also visible.

We will now provision an OCI PostgreSQL Database. OCI Menu > Databases > PostgreSQL > Overview	 The screenshot shows the OCI Cloud console interface. At the top, there is a search bar and a 'Cloud' button. Below the search bar, a sidebar menu lists various services: Home, Compute, Storage, Networking, Oracle AI Database, Databases, Analytics & AI, and Developer Services. The 'Databases' menu item is highlighted with a red box. An arrow points from this box to the 'Overview' link under the 'PostgreSQL' section in the main content area, which is also highlighted with a red box. The 'PostgreSQL' section also lists 'DB Systems' and 'Backups'.
Click Create Database System.	 The screenshot shows the 'Get started with PostgreSQL Database system' page. It features a large heading 'PostgreSQL Database system' and a subheading 'Create a new Database system.' Below this, there is a prominent black button with the text 'Create Database system' highlighted by a red box. To the right of the button is a link labeled 'Learn more'. Below the button, the section 'Configurations' is visible.
Select Create New Database System.	 The screenshot shows the 'Create PostgreSQL Database system' wizard. It has a three-step process: 1. Select creation type, 2. Configure Database system, and 3. Review and create. The first step is active. A red box highlights the 'Create new Database system' option, which includes the description 'Create and configure a new Database system' and a checkmark icon.

Enter a Database System Name. Select PostgreSQL major version as the latest. In my case, version 16.	Database system configuration Database system name in [REDACTED] (Change compartment) Description <i>Optional</i> User-provided data about the Database system. PostgreSQL major version 16
Enter Node Count as 1. Enter Performance tier as 75k IOPS. Select Data Placement as Availability Domain Specific. Select Availability Domain AD-1.	Database system Node count ⓘ 1 One primary node is required. Additional nodes will be reserved as standby/replica nodes. Performance tier ⓘ 75K IOPS Higher performance tiers are available for shapes with 16 or more OCPUs Data placement ☐ Regional Data stored durably in multiple availability domains ☒ Availability Domain-specific Data stored durably within the selected availability domain Availability Domain AD-1 tVMZ:EU-FRANKFURT-1-AD-1 AD-2 tVMZ:EU-FRANKFURT-1-AD-2
Select Shape Series as AMD. Select Shape Filter as Flex. Select Shape name as PostgreSQL.VM.Standard.E5.Flex. Select OCPUs as 2. Select Memory as 16.	Hardware configuration Memory scales with number of OCPU selected. Storage scales depending on usage. Learn more Shape series AMD Flexible OCPU count. Current generation AMD processors. Intel Flexible OCPU count. Current generation Intel processors. Shape filter Flex Flex shapes allow you to configure the amount of OCPU and Memory. Fixed shapes come with set values for OCPU and memory. Shape name PostgreSQL.VM.Standard.E5.Flex OCPUs 2 Input a number between 1 - 64 OCPUs. Memory 16 Input a number between 16 - 128 GB.

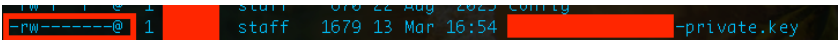
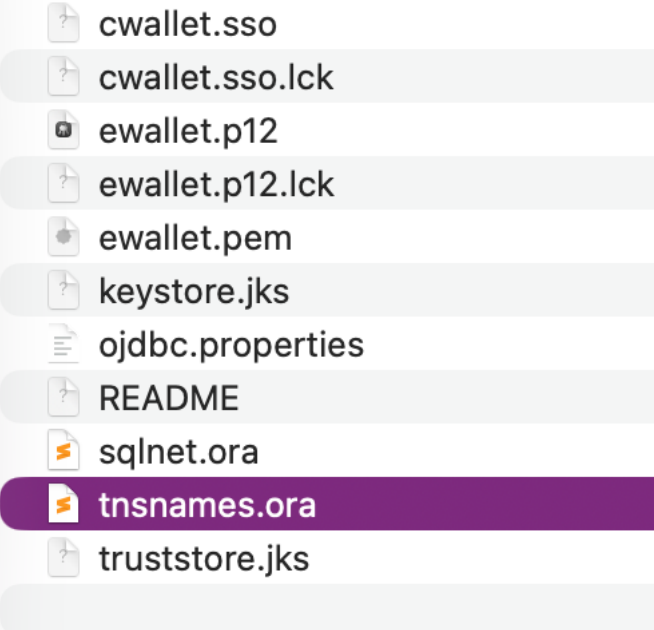
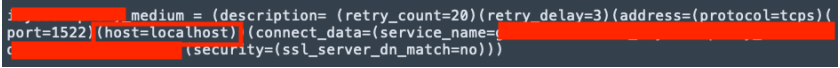
Select the VCN that we created earlier. Select the Private Subnet that we created earlier.	
Enter a Database Username. Select Password Option as Input Password. Enter and Confirm Password. Make Note of them	
Click Submit.	

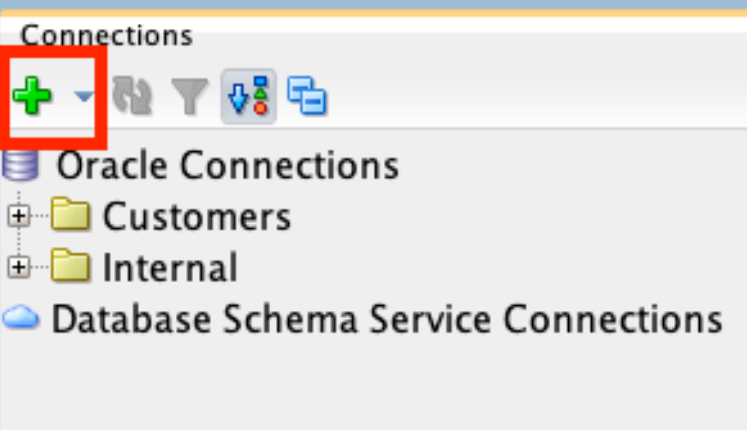
Once your OCI PostgreSQL Service is active. Make note of the Primary endpoint FQDN and Port (5432).	 The screenshot shows the 'Connection details' section of the OCI console. It includes fields for 'Primary endpoint' (redacted), 'Primary endpoint FQDN' (redacted and highlighted with a red box), 'Reader endpoint FQDN' (—), and 'CA certificate' (...IFICATE-----). There are 'Copy' and 'Download' buttons.
Now we will create a Jump Server in order to access our Database Services through an SSH Tunnel. OCI Menu > Compute > Instances.	 The screenshot shows the OCI console navigation menu. The 'Compute' menu item is highlighted with a red box. A red arrow points from the 'Compute' menu item to the 'Instances' sub-menu item, which is also highlighted with a red box. The 'Instances' sub-menu item is located under the 'Compute' section.
Click Create Instance.	 The screenshot shows the 'Instances' page in the OCI console. It includes a search bar, a filter section with 'Applied filters' and 'Compartment' (redacted), and a 'Create instance' button highlighted with a red box. There is also an 'Actions' dropdown menu.

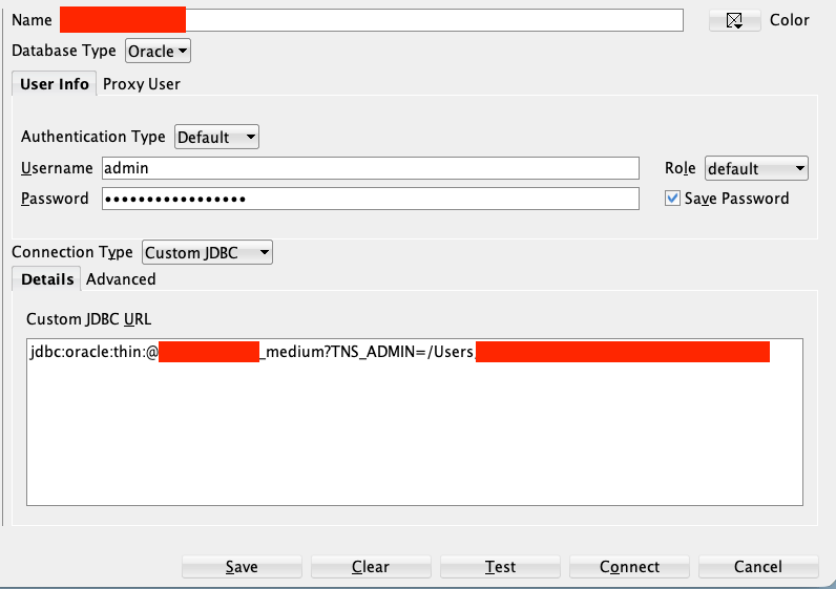
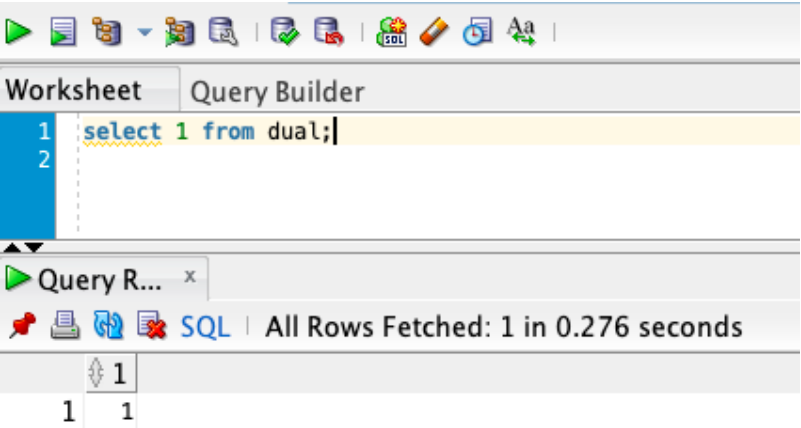
Enter a Name. Select your Compartment. Select the Availability Domain as AD 1.	Create an instance to deploy and run applications, or save as a reusable Terraform Name [Redacted] Create in compartment [Redacted] Placement The availability domain helps determine which shapes are available. Availability domain AD 1 tVMZ:EU-FRANKFURT-1-AD-1 AD 2 tVMZ:EU-F						
Click Change Image. Select the Image as Oracle Linux 9.	Image and shape A shape is a template that determines the number of CPUs, amount of memory, and other resources allocated to an instance. Image Change image  <table> <tr> <td>Operating system</td><td>Oracle Linux 9</td></tr> <tr> <td>Image build</td><td>2026.03.31-0</td></tr> <tr> <td>Security</td><td>Shielded instance, Confidential computing, BM Confidential computing</td></tr> </table>	Operating system	Oracle Linux 9	Image build	2026.03.31-0	Security	Shielded instance, Confidential computing, BM Confidential computing
Operating system	Oracle Linux 9						
Image build	2026.03.31-0						
Security	Shielded instance, Confidential computing, BM Confidential computing						
Create Change Shape. Select the Shape as VM.Standard.E5.Flex with a VM Shape with 1 OCPU and 12 GB Memory.	Shape Change shape  <table> <tr> <td>Shape</td><td>VM.Standard.E5.Flex</td></tr> <tr> <td>Shape build</td><td>Virtual machine, 1 core OCPU, 12 GB memory, 1 Gbps network bandwidth</td></tr> <tr> <td>Security</td><td>Shielded instance, Confidential computing</td></tr> </table> > Advanced options	Shape	VM.Standard.E5.Flex	Shape build	Virtual machine, 1 core OCPU, 12 GB memory, 1 Gbps network bandwidth	Security	Shielded instance, Confidential computing
Shape	VM.Standard.E5.Flex						
Shape build	Virtual machine, 1 core OCPU, 12 GB memory, 1 Gbps network bandwidth						
Security	Shielded instance, Confidential computing						

Leave the Security options as Default.	Security You can enable either shielded instances or confidential computing but not both, simultaneously. Shielded instance ☐ Shielded instances harden the firmware security on bare metal hosts and virtual machines (VMs) to defend against malicious boot level software. Shielded instances use a combination of Secure Boot, Measured Boot, and the Trusted Platform Module (TPM) to harden the firmware security. On some instances, these options must be enabled together. In these cases, when you select one option, any other required options are automatically selected. After a shielded instance is launched, only the name of the instance can be changed. Confidential computing ☐ Confidential computing is hardware technology in CPUs that encrypts data in-use while being processed and protects against these threats. > Advanced options
Select the VCN that we created earlier. Select the Public Subnet that we created earlier.	Primary network ☒ Select existing virtual cloud network ☐ Create new virtual cloud network ☐ Specify OCID Virtual cloud network compartment Virtual cloud network vcn Subnet ☒ Select existing subnet ☐ Create new public subnet Subnet compartment Subnet public subnet
Select the Private IPv4 address as Automatically assign private IPv4 address.	Private IPv4 address assignment If the VNIC must be reachable only by your cloud or on-premises resources, it must have a private private IPv4 addresses. Learn more about private IPv4 addresses Subnet IPv4 prefixes 10.0.0.0/24 Private IPv4 address Automatically assign private IPv4 address Oracle assigns the next unused address in the chosen subnet to the VNIC Manually assign private I Enter an unused address the VNIC

Toggle on <i>Automatically assign public IPv4 address.</i>	Public IPv4 address assignment If the VNIC must be reachable from the internet, it must have a public IP address. Automatically assign public IPv4 address ☒ If you're not sure whether you need a public IP address, you can always assign one later. IPv6 address assignment An IPv6 address can be reachable or not reachable from the internet. Learn more. Assign IPv6 address from subnet prefix ☐								
Select <i>Generate a key pair for me.</i> Click <i>Download Private Key</i> and <i>Download Public Key.</i>	Add SSH keys Generate an SSH key pair to connect to the instance using a Secure Shell (SSH) connection. ☒ Generate a key pair for me ☐ Upload public key file (.pub) ☐ Paste public key ☐ No SSH keys Download the private key so that you can connect to the instance using SSH. Download private key Download public key								
Leave all the <i>Boot Volume settings as Default.</i> Click <i>Create.</i>	Boot volume A boot volume is a detachable device that contains the image used to boot the compute instance. Specify a custom boot volume size and performance setting ☐ Volume performance varies with volume size. Default boot volume size: 46.6 GB. When you specify a custom boot volume size, service limits apply. Use in-transit encryption ☒ Encrypts data in transit between the instance, the boot volume, and the block volumes. Encrypt this volume with a key that you manage ☐ By default, Oracle manages the keys that encrypt this volume, but you can choose a key from a vault that you have access to if you want greater control over the key's lifecycle and how it's used. How do I manage my own encryption keys? Block volumes Attach block volume <table><thead><tr><th>Name</th><th>Attachment type</th><th>Create type </th><th>Reservations</th></tr></thead><tbody><tr><td></td><td></td><td></td><td></td></tr></tbody></table>	Name	Attachment type	Create type	Reservations				
Name	Attachment type	Create type	Reservations						
Once your Instance is created make note of the <i>Public IP Address.</i>	Instance access You connect to a running Linux instance using a Secure Shell (SSH) connection. You'll need the private key from the SSH key pair that was used to create the instance. Public IP address Copy Username opc								

Next, we will change the permissions of our Private Key to only grant Read and Execute Permissions to the user. Using MacOS Terminal or Git Bash for Windows you can run: <i>chmod 600 <name-of-private-key.key></i>	 <pre> 1 staff 076 22 Aug 2023 Config 1 staff 1679 13 Mar 16:54 -private.key </pre>
Unzip our downloaded Autonomous AI Database Wallet File and open the <i>tnsnames.ora</i> file using a text editor in order to change our host to localhost as we will be accessing our Autonomous AI Database via an SSH Tunnel.	 - wallet.sso - wallet.sso.lck - ewallet.p12 - ewallet.p12.lck - ewallet.pem - keystore.jks - ojdbc.properties - README - sqlnet.ora tnsnames.ora - truststore.jks
For one (or all) of your service names, <i>update the host parameter to localhost.</i> <i>Save and close</i> the file.	 <pre> ...medium = (description= (retry_count=20)(retry_delay=3)(address=(protocol=tcps)(port=1522)(host=localhost)(connect_data=(service_name= {security=(ssl_server_dn_match=no)) </pre>

Using MacOS Terminal or Git Bash or PuTTY for Windows we can open up an SSH Tunnel via our Jump Server to connect to our Autonomous AI Database in our Private Subnet. Command for MacOS Terminal or Git Bash is below the screenshot. PuTTY requires configuration via UI.	 <pre>\$ ssh -i <path-to-private-key.key> -N -L localhost:1522:<autonomous-db-private-ip>:1522 opc@<jump-server-public-ip-address></pre> <i>ssh -i <path-to-private-key.key> -N -L localhost:1522:<autonomous-db-private-ip>:1522 opc@<jump-server-public-ip-address></i> Once execute, leave the terminal window or PuTTY session running.
Open your favourite SQL Editor such as SQL Developer Desktop or the VS Code SQL Developer Extension. Create a New Connection.	

Enter a Name. Enter Username as admin. Enter admin user Password. Select Connect Type as Customer JDBC. Enter the Customer JDBC URL following the format shown under the screenshot. Click Test to confirm the connection works. Click Save to save the connection. Click Connect to open a connection to the Autonomous AI Database.	 Custom JDBC URL Format jdbc:oracle:thin:@<db-service-name-from-tnsnames-file>?TNS_ADMIN=<location-of-unzipped-autonomous-wallet-file> Your service name normally takes the format of <database-name>_<service-level> Where service level is either low, medium or high. Depending on the service level you use, ensure you have updated the host parameter to localhost for the corresponding service name in the tnsnames.ora file.
Enter sample query to test database connection works. select 1 from dual;	

For the Autonomous AI Database, we will use the default SH Schema.

For PostgreSQL we will need to load data within database.

In this directory there is a **SALES.csv** file, we will load this into our PostgreSQL.

I have created a Python Notebook that can be executed to load the SALES.csv file within the PostgreSQL database.

You will have to execute this script while within the same Private Subnet as the PostgreSQL database.

I used an OCI Data Science Notebook Session provisioned within the same Private Subnet.

Alternatively, you can convert this Notebook into a .py file and execute it from within your Jump Server we created.

This guide does not cover the steps to load data into PostgreSQL DB.

```
# Establish Connection
conn_string = f'postgresql+psycopg://{DB_USER}:{DB_PASS}@{DB_HOST}:{DB_PORT}/{DB_NAME}'

# Create Engine and Connect
db = create_engine(conn_string)
sa_conn = db.connect()

# Load to PostgreSQL
sales_df.to_sql('sales', con=sa_conn, if_exists= 'replace', index=False)
```

Once the SALES.csv file has been loaded into the PostgreSQL Database we can then configure the Select AI Proxy Integration between the Autonomous AI Database and PostgreSQL Database.

Open and execute the below scripts using the Autonomous AI Database connection we configured within SQL Developer above.

00 - Setup Select AI – Anon.sql

07 - Demo Select AI Proxy Integration – Anon.sql

Setup Select AI - Anon.sql

Demo Select AI Proxy Integration - Anon.sql

End of Technical Guide